

Data-driven dynamic modeling and control of pneumatic soft actuators with stability analysis

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Soft actuators enable safe interaction, compliance, and adaptability in unstructured environments, but their distributed, highly nonlinear dynamics make accurate modeling and control difficult. To address these challenges, we present a data-driven modeling and control framework for a planar pneumatic soft actuator that enables stable closed-loop tip tracking without requiring an explicit first-principles model. The approach decomposes the system into (i) a sparse, nonlinear forward map that captures the dominant static input–output relationship and provides feedforward actuation, and (ii) a compact extended dynamic mode decomposition with control (EDMDc) residual dynamics model that captures transient behavior. A central contribution is an accompanying stability analysis: by embedding the learned residual dynamics and feedback controller into an augmented linear closed-loop system, we can predict closed-loop local stability directly (e.g., via the spectral radius of the augmented dynamics). We then experimentally validate these stability predictions, showing stable behavior in the regimes identified by the analysis. We also demonstrate accurate tip tracking on slow, shape-rich trajectories with approximately 2 mm root mean squared error (RMSE) for 10 mm/s references and stable, reliable tracking at substantially higher bandwidths (below 10 mm RMSE for 100 mm/s references). These results demonstrate that structured, control-oriented learning can deliver deployable performance together with predictive stability insight, motivating future extension to spatial soft actuators where analytical modeling is especially challenging.

Keywords: dynamic modeling, data-driven, task-space control, soft actuator

1 Introduction

Soft actuators offer capabilities that complement—and in some settings surpass—those of rigid systems [1]. Their low stiffness and mass enable safe physical interaction with humans and fragile objects, while their distributed compliance allows them to conform to unstructured environments [2]. These features make soft robotic systems attractive for surgical tasks, delicate grasping, and adaptive locomotion, where contact-rich interaction and tolerance to uncertainty are essential [3–5]. However, realizing this in practice relies on control strategies that can deliver accurate, repeatable motion despite substantial body deformation and strong coupling between actuation and geometry.

Modeling and controlling soft actuators is challenging because their deformation is fundamentally distributed: rather than being well-approximated by a small set of joint coordinates, soft bodies exhibit continuum mechanics with effectively infinite degrees of freedom. The resulting dynamics are typically nonlinear, history dependent, and sensitive to manufacturing variability, external loading, and environmental contact [6]. Pneumatically actuated soft actuators in particular, have additional complexity from pressure dynamics, valve and flow nonlinearities, viscoelastic material behavior, and rate-dependent effects that can introduce hysteresis and drift [7,8]. These factors can make first-principles models difficult to derive, time-consuming to calibrate, and sensitive to operating conditions.

Existing approaches to soft-robot modeling for control broadly fall into two categories. The first includes reduced-order models, which seek tractable dynamics by introducing structure (e.g., piecewise-constant curvature assumptions [9], Cosserat-rod formulations [10], or finite-element reductions [11]) and by lumping distributed effects into a small number of generalized coordinates

and parameters. These models produce interpretable controllers and provide useful physical insight, but often require restrictive assumptions, significant system-specific identification effort, and may only be reliable within a limited operating envelope—especially when large deformations, rapid motions, or complex contact conditions are present.

The second category comprises data-driven and neural-network (NN) models that learn input–output mappings directly from experimental data. NN-based models can be highly expressive and accurate predictors, but they commonly require substantial training data and computational resources, and their nonlinear structure can complicate controller synthesis, stability analysis, and real-time deployment [12–14]. These considerations have led to continued interest in data-driven modeling approaches that retain computational simplicity for control while capturing the dominant nonlinear behaviors of soft pneumatic actuators.

Motivated by these tradeoffs, we propose a middle ground approach: a structure-preserving data-driven architecture that isolates the dominant geometric nonlinearity from the transient dynamics that are amenable to linear approximation. Our framework decomposes the soft actuator behavior into (i) a static nonlinear kinematic mapping and (ii) a dynamic linear correction. The static mapping is identified from steady-state data using sparse polynomial regression and is inverted to generate feedforward input. The residual transient dynamics around this static manifold are modeled using an extended dynamic mode decomposition with control (EDMDc) approach [15,16], i.e., a discrete-time linear state-space model formed with time-delay observables. This decomposition preserves interpretability, produces a control-oriented linear model for feedback synthesis, and enables stability analysis.

Prior work has demonstrated that data-driven modeling frameworks such as SINDYc [17] and Koopman [18]/EDMD representations enable nonlinear or optimal control of soft actuators. In particular, sparse regression combined with sliding-mode control

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has been used to identify full nonlinear dynamics and design robust nonlinear feedback laws for a cable-driven actuator, although primarily in quasi-static motion regimes [19]. Koopman-based approaches have enabled optimal control of soft pneumatic actuators in the inertial regime through globally linear lifted representations and LQR design [20]. These methods demonstrate high-speed, large-deflection tracking performance, but require full-state optimal control in the lifted space and assume sufficient controllability of the learned linear system.

In contrast, our approach does not attempt to identify or control the full nonlinear dynamics, which may not be possible for dynamic trajectory following tasks. Furthermore, pneumatic soft actuators are generally underactuated in task-space, which makes it challenging for full-state optimal control design. Our formulation circumvents this issue by directly regulating task-space error through model-based decoupling. The resulting architecture achieves dynamic trajectory tracking while remaining computationally simple and tunable for practical deployment.

The main contributions of this work are:

- **A control-oriented static–dynamic decomposition for soft actuators.** We combine a sparse polynomial steady-state map with a delay-lifted EDMDC residual model to obtain a compact linear state-space representation. The resulting dynamic model enables the use of standard linear control tools (e.g., PI design and eigenvalue-based stability analysis) without requiring full nonlinear controller synthesis.
- **Accurate tracking across quasi-static and dynamic regimes.** Hardware experiments on a 3D printed soft pneumatic actuator demonstrate precise slow-trajectory tracking (≈ 2 mm RMSE) and stable dynamic tracking up to 0.5 Hz (tip speeds > 100 mm/s) with < 10 mm RMSE.
- **Model-based prediction of closed-loop local stability.** We construct an augmented closed-loop system from the learned model and show that its spectral radius predicts tracking behavior by distinguishing stable, marginally oscillatory, and unstable regimes prior to hardware deployment.

The rest of this article is structured as follows: Section 2 presents both the kinematic and dynamic models, Section 3 introduces hardware considerations, including soft actuator design and experiment setup. Section 4 presents experiment results including stability analysis and trajectory tracking results. We conclude with a discussion in Section 5 and considerations for future work in Section 6.

2 Model

This section presents the learned kinematic map, residual dynamic model, and resulting control law. The input $u \in \mathbb{R}^{n_u \times 1}$ is linearly scaled to reference pressures for each of the four chambers. Here we consider $n_u = 2$ (i.e., $u = [u_1, u_2]^T$ and $u_i \in [-0.02, 0.02]$) and negative values of u_1, u_2 represent antagonistic pressurization (e.g., $-u_1$ and $+u_1$ represent pressurization of the back left and back right chambers, respectively). The output is the tip position of the planar soft actuator $p = [x_{tip}, y_{tip}]^T \in \mathbb{R}^{2 \times 1}$.

2.1 Kinematic feedforward map. A sparse, kinematic forward model is fit from pressure to tip position such that:

$$p \approx W^T \phi(u), \quad (1)$$

where $W \in \mathbb{R}^{q \times 2}$ is the learned coefficient matrix and $\phi(u) \in \mathbb{R}^{q \times 1}$ is the basis function library. We consider a set of monomials up to degree $d = 2$, resulting in $\phi(u) = [1, u_1, u_2, u_1^2, u_2^2, u_1 u_2]^T$. This choice of basis functions was motivated by the empirically observed curvature in the actuator's shape. This is consistent with the sparse-identification results in [19], where the polynomial/monomial terms were retained as the dominant contributors.

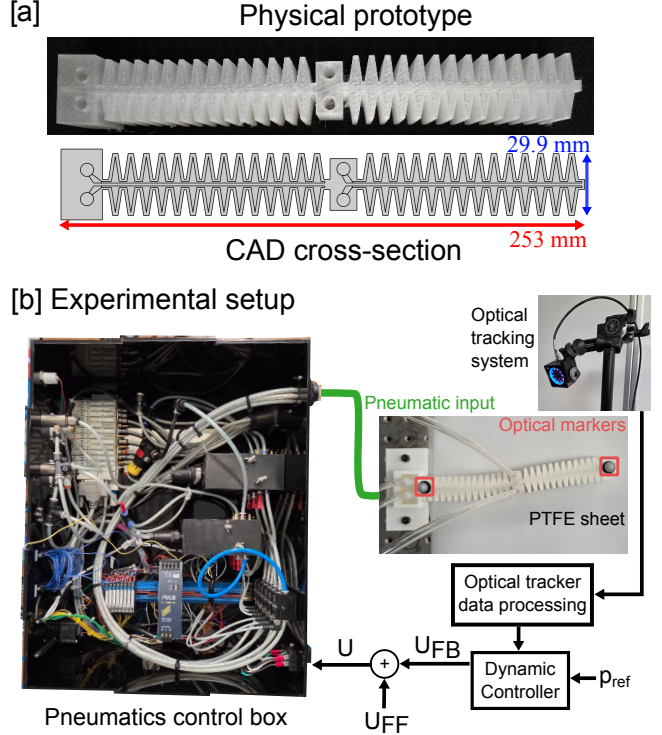


Fig. 1 The soft actuator used in this work is a four chamber PneuNet (a). The custom pneumatics box used for experimentation includes proportional spool valves and pressure transducers for independent chamber actuation. The actuator tip position is measured by an optical tracker and used to compute the feedback input u_{FB} . The total command $u = u_{FB} + u_{FF}$ is then saturated and converted to desired chamber pressures (b).

To generate the dataset for this map, u was chosen to be staircase functions with a period of 3 s for a total of 330 s. We build static maps from u to p by associating a tip position p based on the mean position of the last 0.5 s of each period for a corresponding input u resulting in a total of 110 configurations in the workspace. We then construct $\Phi = [\phi_1^T; \dots; \phi_m^T] \in \mathbb{R}^{m \times q}$ and $P = [p_1^T; \dots; p_m^T] \in \mathbb{R}^{m \times 2}$ to find W such that $P \approx \Phi(u)W$. The regularized least-squares solution is given by:

$$W = (\Phi^T \Phi + \lambda I)^{-1} \Phi^T P, \quad (2)$$

where $\lambda = 1 \times 10^{-6}$ is the regularization parameter. This resulted in the following map from input u to estimated tip position $(\hat{x}_{tip}, \hat{y}_{tip})$:

$$\hat{x}_{tip} = -0.0197 + 14.1u_1 + 5.70u_2 + 165u_1^2 + 91.2u_2^2, \quad (3)$$

$$\hat{y}_{tip} = 0.208 + 0.314u_1 + 0.384u_2 - 805u_1^2 - 533u_2^2 - 774u_1u_2, \quad (4)$$

which yielded an RMSE of 2.96 cm and $R^2 = 0.92$ for $\hat{x}_{tip}$, and an RMSE of 1.34 cm and $R^2 = 0.94$ for $\hat{y}_{tip}$. The larger error in $\hat{x}_{tip}$ was attributed to asymmetric bending, likely due to manufacturing nonuniformity, and was most pronounced near the workspace boundary. Although increasing the polynomial degree reduced this error (e.g., to 1.9 cm for $d = 3$), $d = 2$ was retained to preserve sparsity, with the residual dynamic model compensating for much of the remaining mismatch in closed-loop operation.

Therefore, for a desired reference at time k , $p_{ref,k} \in \mathbb{R}^{2 \times 1}$, the corresponding feedforward input $u_{FF,k} \in \mathbb{R}^{2 \times 1}$ is obtained by

numerically inverting the static mapping. We solve the bounded nonlinear problem $\arg\min_u H(u)$ with $u_{min} \leq u \leq u_{max}$ where:

$$H(u) = \|p_{ref,k} - W^T \phi(u)\|_2^2 + \lambda_u \|u\|_2^2 + \lambda_{\Delta u} \|u - u_{k-1}\|_2^2, \quad (5)$$

using *fmincon* in MATLAB 2024b. The parameters $\lambda_u = 10$, $\lambda_{\Delta u} = 10$ were selected empirically to balance the tracking, input-magnitude, and smoothness terms in the cost function $H(u)$. This was applied pointwise along the reference trajectory, $p_{ref}(t)$ to generate $u_{FF}(t)$.

2.2 Dynamic model. The dynamic model identification approach is based on residual EDMDc with time delays. The general idea is to transform the system coordinates to linear, lifted coordinates and identify the residual system dynamics.

Using the inputs from the same dataset described in 2.1, the feedforward predicted tip position, $p_{FF} = [\hat{x}_{tip}, \hat{y}_{tip}]^T$ is computed. The residual at time k , $r_k = p_k - p_{FF,k}$ is used to construct the lifted state:

$$z_k = [r_k, r_{k-1}, \dots, r_{k-t_r}]^T \in \mathbb{R}^{2(t_r+1) \times 1}, \quad (6)$$

where t_r is the residual time history. We found that $t_r = 2$ was sufficient to capture the dominant history associated with the actuator and pneumatic dynamics. These entries are stacked as matrices (let $n_z = 2(t_r + 1)$):

$$Z_1 = [z_1, \dots, z_K] \in \mathbb{R}^{n_z \times K} \quad (7)$$

$$Z_2 = [z_2, \dots, z_{K+1}] \in \mathbb{R}^{n_z \times K} \quad (8)$$

$$U = [u_1, \dots, u_K] \in \mathbb{R}^{n_u \times K} \quad (9)$$

The training set is normalized; we calculate per-component means, $\mu_Z \in \mathbb{R}^{n_z \times 1}$, $\mu_U \in \mathbb{R}^{n_u \times 1}$ and diagonal standard deviation matrices, $S_z \in \mathbb{R}^{n_z \times n_z}$, $S_u \in \mathbb{R}^{n_u \times n_u}$, such that $\tilde{Z}_1 = S_z^{-1}(Z_1 - \mu_Z \mathbf{1})$, where $\mathbf{1} \in \mathbb{R}^{1 \times K}$ denotes the all-ones vector. The dynamic model is discrete-time linear on the normalized lifted coordinates and is given by:

$$\tilde{z}_{k+1} = A\tilde{z}_k + B\tilde{u}_k, \quad (10)$$

where $A \in \mathbb{R}^{n_z \times n_z}$ and $B \in \mathbb{R}^{n_z \times n_u}$. We apply the standard EDMDc process and construct:

$$\Omega = \begin{bmatrix} \tilde{Z}_1 \\ \tilde{U} \end{bmatrix} \in \mathbb{R}^{(n_z+n_u) \times K}, \quad (11)$$

$$G = \begin{bmatrix} A & B \end{bmatrix} \in \mathbb{R}^{n_z \times (n_z+n_u)}, \quad (12)$$

to set up the problem:

$$\tilde{Z}_2 = G\Omega \quad (13)$$

Once again using ridge regression, the optimal (in the least-squares sense) solution is given by:

$$G = \tilde{Z}_2 \Omega (\Omega \Omega^T + \lambda I)^{-1} \quad (14)$$

The learned (A, B) is used for online control as described next. The output map (C) is used to extract the denormalized first residual block:

$$C = \begin{bmatrix} I_2 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times n_z}, \quad (15)$$

$$r_k = C(S_z \tilde{z}_k + \mu_z), \quad (16)$$

The reconstructed predicted tip position $\hat{p}$ is the sum of the feedforward map output $p_{FF,k}$ and the estimated residual $\hat{r}_{k+1}$:

$$\hat{p}_{k+1} = p_{FF,k+1} + \hat{r}_{k+1}. \quad (17)$$

This can be used online to predict the next tip position; however, in the experiments reported here, feedback is computed from the measured tip position rather than this one-step-ahead prediction.

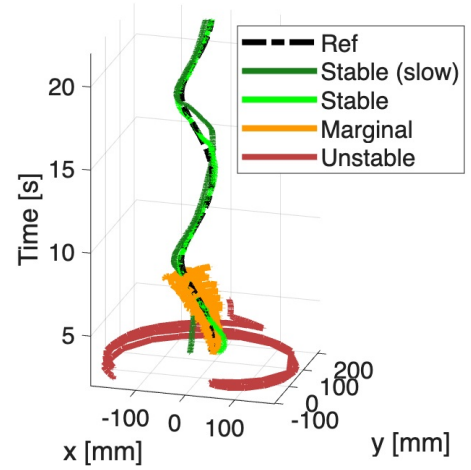


Fig. 2 Experimental tip trajectories in task space over time (x-y-t) for a 0.1 Hz circular reference signal (dashed). Prior to each experiment, the proposed augmented linear stability test was used to predict the closed-loop stability of the selected controller settings from a random gain sweep. Colored traces show the resulting experimental responses, which agree well with these predictions: green indicates predicted stable behavior (including a conservative “slow” setting), orange denotes marginal stability, and red denotes unstable behavior. The last two tests were terminated early to prevent actuator damage.

2.3 Controller. To determine the feedback input u_{FB} , we consider the mapping from input to output space (e.g., $\Delta p = J\Delta u$) and invert this mapping for a desired change in tip position (i.e., $\Delta u_{FB} = J^T (JJ^T + \alpha I)^{-1} \Delta p_{des}$). We construct this mapping J by considering the change in the normalized lifted state to a normalized input change $\Delta \tilde{z} = B\Delta \tilde{u}$. Denormalizing both sides leads to $\Delta z = S_z B S_u^{-1} \Delta u$. Substituting $\Delta p = C\Delta z$ leads to:

$$J = C S_z B S_u^{-1}. \quad (18)$$

Let $E = \int edt$ and the error $e = p_{ref} - p$. We consider the desired effort $p_{des} = K_P e + K_I E$, where $K_P, K_I \in \mathbb{R}^{2 \times 2}$ are diagonal gain matrices, resulting in:

$$u_{FB,k} = J^T (JJ^T + \alpha I)^{-1} (K_P e_k + K_I E_k). \quad (19)$$

The final output is

$$u_k = u_{FB,k} + u_{FF,k}, \quad (20)$$

which is saturated and sent to pressure valve command.

3 Hardware

3.1 Pneumatic soft actuator design. The experimental platform is a planar pneumatic soft actuator (Fig. 1(a)) based on the PneuNet architecture [21], where pressurization causes asymmetric expansion in compliant regions to generate planar bending. The body contains four independently pressurized chambers, enabling bidirectional curvature control via differential pressurization. The actuator was 3D printed on a Bambu Lab X1C using TPU (Polymer PolyFlex TPU 90A) with a uniform 1.5 mm wall thickness. Each chamber comprises 14 triangular segments; overall dimensions are 253 mm length, 29.9 mm width, and 27.5 mm height.

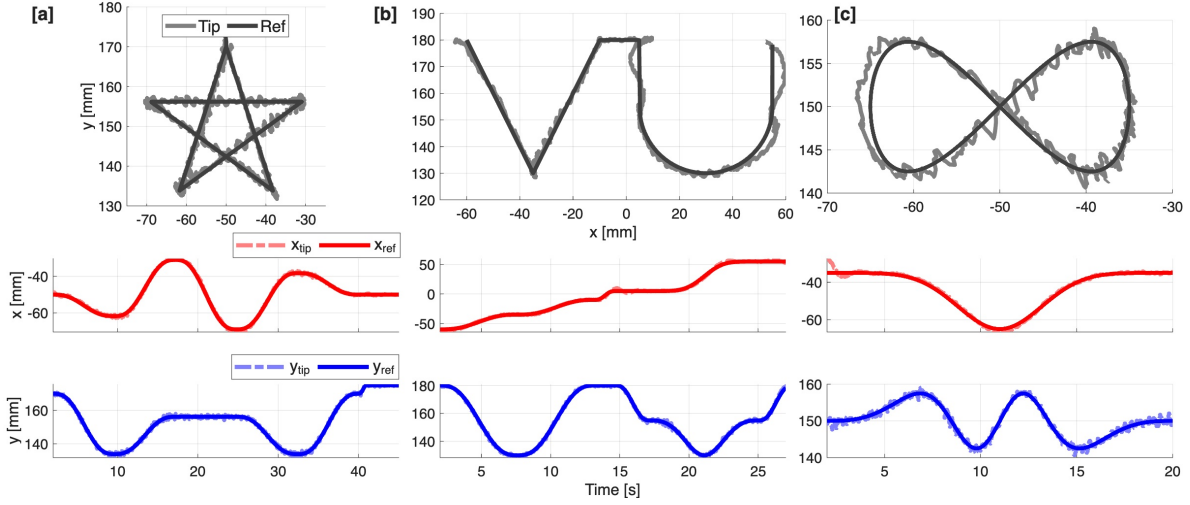


Fig. 3 Precise shape tracking: Averaged tracking performance over five trials for three task-space references: (a) five-point star (43 s, 5 mm/s; RMSE = 1.79 mm), (b) “VU” (25 s, 10 mm/s; RMSE = 2.52 mm), and (c) lemniscate (18 s, average 5 mm/s, peak 15 mm/s; RMSE = 1.98 mm). Top row: task-space paths (mean tip and reference). Middle (x) and bottom (y) rows: tip-displacement components for the mean tip and reference.

3.2 Experimental setup. The actuator’s four chambers are independently driven by a custom pneumatic control box (Fig. 1(b)). The chambers are connected through 4 mm ID / 6 mm OD tubing (McMaster-Carr 50315K69) with barbed fittings (McMaster-Carr 5463K628). The actuator operates over a PTFE sheet (McMaster-Carr 9266K21) to reduce friction and is mounted to a mechanical breadboard using a custom 3D-printed fixture.

The control box houses four 5-port/3-way proportional spool valves (Festo MPYE-5-M5-010-B) with inline pressure sensors (Festo SDE5-D10-NF-T14-V-M8). A mixed I/O DAQ (Humusoft MF634) interfaces the hardware to a desktop PC (AMD Ryzen 7 5700G, 3.80 GHz) running MATLAB 2024b Simulink Desktop Real-Time, enabling independent closed-loop pressure control for each chamber.

The soft actuator is instrumented with two optical markers - the base marker sets the origin and the tip position is used for tracking. Motion is captured with an OptiTrack Prime 13 system streaming Cartesian position at 50 Hz. Valve commands and the dynamic model in Eq. (10) run at 50 Hz, while pressure sensing runs at 1 kHz.

4 Results

4.1 Stability analysis. Here we derive the closed-loop system to determine conditions for stability. The procedure is to rewrite the closed-loop system dynamics in terms of the normalized state, $\tilde{z}_k$, construct an augmented state transition matrix A_{aug} and ensure that its spectral radius is less than 1.

The lifted linear dynamics is given by Eq. (10), where the lifted input at time k is given by $\tilde{u}_k = \tilde{u}_{FF,k} + \tilde{u}_{FB,k}$. First, we express $u_{FB,k}$, given in Eq. (19), using normalized input coordinates. Let $\tilde{K}_P = J^T (JJ^T + \alpha I)^{-1} K_P$ and $\tilde{K}_I = J^T (JJ^T + \alpha I)^{-1} K_I$. Therefore,

$$u_{FB,k} = \tilde{K}_P e_k + \tilde{K}_I E_k. \quad (21)$$

Note that no centering term appears in the normalized feedback input, since the input mean μ_u is already accounted for in $\tilde{u}_{FF,k}$. Therefore, the normalized feedback term is written as:

$$\tilde{u}_{FB,k} = S_u^{-1} \tilde{K}_P e_k + S_u^{-1} \tilde{K}_I E_k. \quad (22)$$

Substituting Eq. (22) into the dynamics given in Eq. (10) leads to:

$$\tilde{z}_{k+1} = A\tilde{z}_k + B\tilde{u}_{FF,k} + BS_u^{-1} \tilde{K}_P e_k + BS_u^{-1} \tilde{K}_I E_k. \quad (23)$$

The error e_k can be rewritten in terms of $\tilde{z}_k$ using Eq. (16) as follows:

$$e_k = p_{ref,k} - p_k, \quad (24)$$

$$= p_{ref,k} - (PFF_{k,k} + r_k), \quad (25)$$

$$= p_\mu - CS_z \tilde{z}_k, \quad (26)$$

where $p_\mu = p_{ref,k} - PFF_{k,k} - CS_z \mu_z$. Substituting Eq. (26) in Eq. (23):

$$\tilde{z}_{k+1} = A\tilde{z}_k + B\tilde{u}_{FF,k} + BS_u^{-1} \tilde{K}_P (p_\mu - CS_z \tilde{z}_k) + BS_u^{-1} \tilde{K}_I E_k, \quad (27)$$

$$= (A - BS_u^{-1} \tilde{K}_P CS_z) \tilde{z}_k + BS_u^{-1} \tilde{K}_I E_k + p_\alpha, \quad (28)$$

where $p_\alpha = B\tilde{u}_{FF,k} + BS_u^{-1} \tilde{K}_P p_\mu$ contains exogenous terms. Define an augmented state vector $q_k = [\tilde{z}_k; E_k] \in \mathbb{R}^{(nz+2) \times 1}$ and note $E_{k+1} = E_k + e_k = E_k + p_\mu - CS_z \tilde{z}_k$ from Eq. (26). Therefore, the augmented system dynamics are given by:

$$q_{k+1} = \begin{bmatrix} A - BS_u^{-1} \tilde{K}_P CS_z & BS_u^{-1} \tilde{K}_I \\ -CS_z & I \end{bmatrix} q_k + \begin{bmatrix} p_\alpha \\ p_\mu \end{bmatrix}, \quad (29)$$

$$= A_{aug} q_k + g. \quad (30)$$

Here $g \in \mathbb{R}^{(nz+2) \times 1}$ contains exogenous terms that do not affect stability. The closed-loop stability is governed only by the spectral radius of $A_{aug} \in \mathbb{R}^{(nz+2) \times (nz+2)}$; stability is predicted when $\rho(A_{aug}) < 1$.

We experimentally evaluated this result by sampling random sets of PI diagonal gains K_P and K_I , constructing the corresponding augmented system A_{aug} , computing its spectral radius $\rho(A_{aug})$, and executing the controller. Figure 2 summarizes the observed behavior across these regimes for a 0.1 Hz circular tracking task. As predicted, system performance changes sharply near the stability boundary $\rho(A_{aug}) = 1$. We started with conservative gains and gradually increased the magnitude and variance of the random gains until we approached the boundary. For relatively low gains ($K_{P,i} \sim 1e-5, K_{I,i} \sim 1e-4$), the system remains stable but exhibits larger tracking error (dark green, $\rho(A_{aug}) = 0.9993$). Larger magnitude gains ($K_{P,i} \sim 1e-4, K_{I,i} \sim$

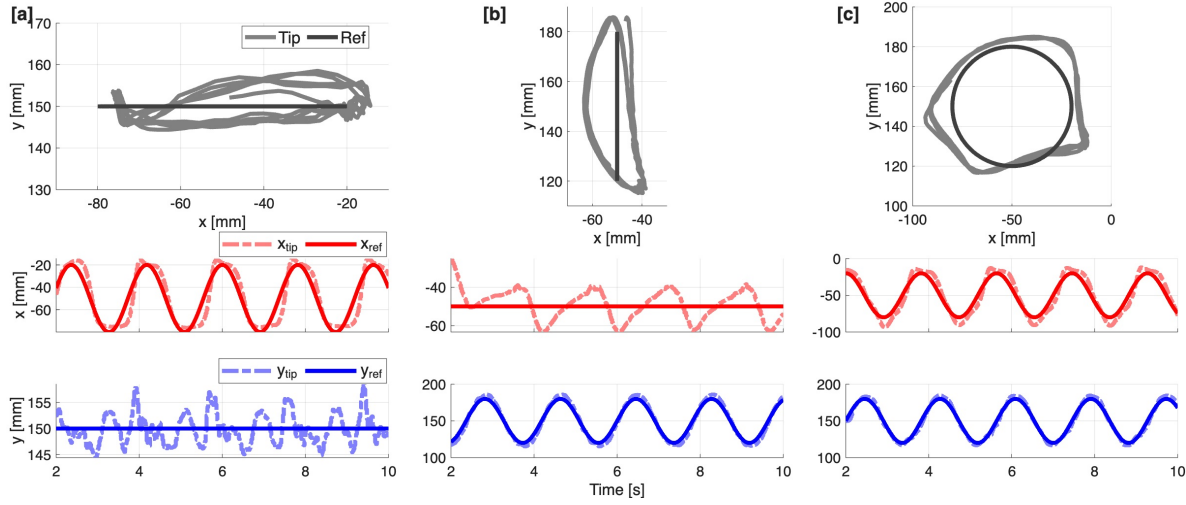


Fig. 4 Higher bandwidth tracking: Averaged tracking performance over five trials for three references at 0.55 Hz: (a) 60 mm horizontal line (RMSE = 9.54 mm), (b) 60 mm vertical line (RMSE = 9.33 mm), and (c) 60 mm-diameter circle (RMSE = 9.41 mm). Top row: task-space paths (mean tip and reference). Middle (x) and bottom (y) rows: tip-displacement components for the mean tip and reference. The trajectories correspond to a peak reference tip speed of 104 mm/s with peak acceleration of 358 mm/s².

1e-3) improve tracking accuracy while maintaining stability (light green, $\rho(A_{aug}) = 0.9998$). As the boundary is approached ($K_{P,i} \sim 1e-4$, $K_{I,i} \sim 2e-3$), the response becomes increasingly oscillatory (orange, $\rho(A_{aug}) = 0.9999$), and we terminate the experiment early to prevent actuator damage. When the boundary is exceeded ($K_{P,i} \sim 2e-4$, $K_{I,i} \sim 2e-3$), the closed-loop system becomes unstable and diverges rapidly (red, $\rho(A_{aug}) = 1.005$), again requiring early termination. These results demonstrate that the augmented learned linear model can reliably predict the local closed-loop behavior across stable, marginal, and unstable regimes.

4.2 Tracking results. We evaluate the controller on a set of low- and high-speed trajectories to assess performance across a broad range of bandwidths. In all trials, the first 2 s are used to drive the tip from its unactuated pose to the initial point of the reference.

Figure 3 summarizes the slow-trajectory results for three tasks: (a) tracing a five-point star over 43 s (5 mm/s), (b) writing “VU” over 25 s (10 mm/s), and (c) tracking a lemniscate over 18 s (average ≈ 5 mm/s, peak ≈ 15 mm/s). Each experiment is repeated five times: the top row shows the averaged task-space paths (reference and mean tip), while the bottom row shows the averaged tip-displacement components (x and y). The resulting mean RMSE values are 1.79 mm for (a), 2.52 mm for (b), and 1.98 mm for (c).

Motivated by these results, we next evaluate high-speed tracking performance (Fig. 4). We consider three trajectories at 0.55 Hz: (a) a 60 mm horizontal line, (b) a 60 mm vertical line, and (c) a 60 mm-diameter circle. This corresponds to a reference tip speed of 104 mm/s with peak acceleration of 358 mm/s², approximately an order of magnitude higher than the slow-trajectory tests. Each task is repeated five times; the top row of Fig. 4 shows the averaged task-space motion and the bottom row shows the corresponding tip-displacement traces. The mean RMSE values are 9.54 mm for (a), 9.33 mm for (b), and 9.41 mm for (c).

5 Discussion

A key motivation for the proposed static–dynamic decomposition is to retain the practical advantages of linear control—transparent tuning, low computational overhead, and analyzable closed-loop behavior—while still capturing the dominant nonlinearities inherent to pneumatic soft actuators. By learning a sparse static map for the input–output nonlinearity and reserving

EDMDC for the transient residual dynamics, the controller avoids attempting to model the full nonlinear continuum dynamics directly, yet still leverages a control-oriented linear representation where it matters most for tracking.

5.1 Stability analysis as a design tool for data-driven controllers. One of the main practical challenges in data-driven soft-robot control is that highly expressive learned models (e.g., black-box NNs) can make stability difficult to assess prior to deployment. In contrast, the residual EDMDC model enables an explicit closed-loop construction with PI feedback, yielding an augmented linear system whose stability can be predicted by a simple spectral-radius condition, $\rho(A_{aug}) < 1$.

This is valuable in two ways. First, it provides a principled way to tune gains (and regularization/pseudoinverse parameters) while explicitly accounting for the learned dynamics, rather than relying on purely empirical trial-and-error. Second, it creates a lightweight “sanity check” that can flag potentially unstable regimes before hardware testing—an important consideration for soft pneumatic systems where saturation, valve nonlinearities, and viscoelastic drift can rapidly lead to limit cycles or divergence if feedback is too aggressive.

This perspective aligns with a broader theme in the literature: using operator-theoretic or sparse identified models to recover enough structure for analysis and controller synthesis, rather than aiming for perfect prediction fidelity. Koopman-based soft-arm control, for example, uses linear lifted dynamics to enable LQR-type designs, emphasizing computational efficiency and deployability. Likewise, SINDYc-based approaches emphasize parsimonious models that can support theoretical stability arguments and robust designs.

5.2 Tracking experiments. The tracking experiments demonstrate a clear regime-dependent behavior: millimeter-scale accuracy on slow, shape-rich references (star/VU/lemniscate) and centimeter-scale errors on high-bandwidth periodic motion (0.55 Hz, 60 mm trajectories). In particular, the VU writing task required substantial curvature variation ($x = -60$ to 50 mm), including motion toward and past the unactuated equilibrium configuration of the soft actuator ($y=0$).

This is consistent with the architecture’s intent. When motion is slow, the static inverse map accounts for most of the input needed to realize the desired pose, and the residual dynamics remain small;

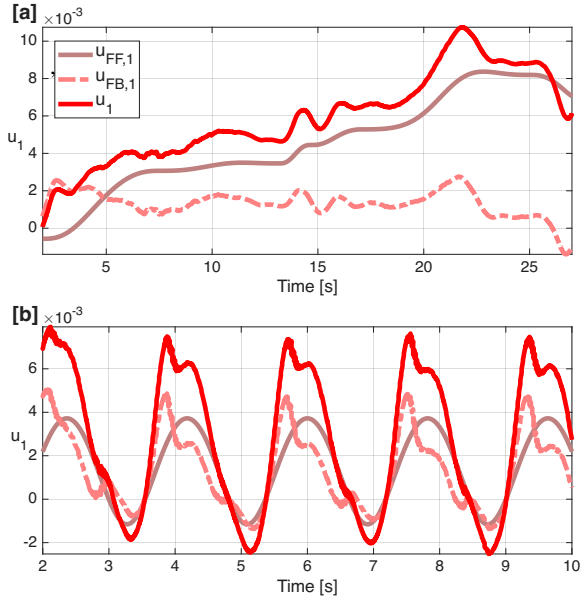


Fig. 5 The input u_1 is decomposed into the individual feedforward ($u_{FF,1}$) and feedback ($u_{FB,1}$) components for the (a) slower VU tracing (10 mm/s) and the (b) faster 0.55 Hz horizontal line (peak 104 mm/s) tracking tasks. The relative contribution (based on integrated absolute magnitudes) of $u_{FB,1}$ to u_1 increases from (a) 23.1% to (b) 46.6%, representing a larger required feedback correction during dynamic motion.

the EDMDC correction and PI loop mainly compensate for modest transient effects and steady-state biases. As tip speed increases, inertial and pneumatic flow/pressure dynamics become more prominent, and the residual model must explain a larger fraction of the motion. To visualize and quantify this effect, we plot the entire u_1 input command for a representative slow- (VU, 10 mm/s) and high-speed (0.55 Hz horizontal line, average ≈ 66 mm/s, peak ≈ 104 mm/s) tracking task in Fig. 5. The feedback component, $u_{FB,1}$, remains modest and roughly constant during the slow VU task. In contrast, it is somewhat periodic for the faster line tracking task with its peak magnitude cyclically surpassing $u_{FF,1}$. To quantify this, the ratio $\frac{\int_0^T |u_{FB,1}(t)| dt}{\int_0^T (|u_{FF,1}(t)| + |u_{FB,1}(t)|) dt}$ is computed and reflects the relative integrated absolute contribution of the feedback component to the overall signal. We find that this contribution doubles from 23.1% to 46.6% between the slow VU and 0.55 Hz line task.

5.3 Limitations. A primary advantage of the proposed data-driven formulation is that it does not require deriving an explicit first-principles model, which becomes increasingly difficult for spatial (3D) soft actuators with inertial effects, complex deformation modes, and strong actuation nonlinearities. This makes the approach well-suited to extending beyond planar platforms, consistent with prior data-driven soft-robot control studies in 3D settings. In this work, however, we validate the method only on a planar actuator, and extending the same pipeline to 3D soft manipulators is a natural next step.

In addition, all experiments focus on free-space tracking. Generalizing the method to tasks involving contact would require accounting for environmental interaction forces, unilateral constraints, and potentially discontinuous dynamics, which are not captured by the present modeling and control structure. As a result, while the current framework is effective for reference tracking in unconstrained motion, accommodating interaction-rich behav-

iors remains an important direction for future work.

6 Conclusion

In this work, we present a data-driven modeling and control framework for a planar pneumatic soft actuator that combines a sparse static forward map with an EDMDC residual dynamics model to enable stable, low-complexity closed-loop tip tracking. The accompanying stability analysis provides a practical criterion for controller design and gain tuning directly in terms of the learned linear augmented dynamics. Experiments demonstrate accurate tracking on slow, shape-rich trajectories (2 mm RMSE for 10 mm/s references) and stable, reliable tracking at substantially higher bandwidths (< 10 mm RMSE for 100 mm/s references). These results suggest that structured, control-oriented learning can deliver deployable performance without requiring explicit first-principles dynamics, and motivate our next step of extending the approach to spatial soft actuators where modeling is especially challenging.

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